

Logistic Regression & Gaussian Discriminant Analysis

A Complete Worked Example: University Admissions

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1 The Problem and Dataset

Task. Predict whether a student will be *admitted* ($y = 1$) or *rejected* ($y = 0$) to a university programme, based on two exam scores.

Feature Vector

For each student i :

$$\mathbf{x}^{(i)} = \begin{pmatrix} x_1^{(i)} \\ x_2^{(i)} \end{pmatrix} \in \mathbb{R}^2, \quad x_1 = \text{Exam 1 score}, \quad x_2 = \text{Exam 2 score}, \quad y \in \{0, 1\}.$$

1.1 Training Data

We have $N = 5$ training examples.

Student	Exam 1 (x_1)	Exam 2 (x_2)	Admitted? (y)
1	45	85	1 (Yes)
2	85	46	1 (Yes)
3	60	83	1 (Yes)
4	30	49	0 (No)
5	68	28	0 (No)

Class summary: **3 admitted** ($y = 1$: students 1, 2, 3) and **2 rejected** ($y = 0$: students 4, 5).

1.2 Test Point

After fitting both models we will classify:

$$\mathbf{x}^* = \begin{pmatrix} 45 \\ 85 \end{pmatrix} \quad (\text{Exam 1} = 45, \text{ Exam 2} = 85).$$

3 Gaussian Discriminant Analysis (GDA)

3.1 Model

GDA is a *generative* model. It assumes:

$$\begin{aligned}y &\sim \text{Bernoulli}(\phi), \\ \mathbf{x} \mid y = 0 &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}), \\ \mathbf{x} \mid y = 1 &\sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}),\end{aligned}$$

with a **shared** covariance $\boldsymbol{\Sigma}$ for both classes. Classification uses Bayes' rule:

$$\hat{y} = \arg \max_{y \in \{0,1\}} P(y \mid \mathbf{x}) = \arg \max_y P(\mathbf{x} \mid y) P(y).$$

3.2 MLE Parameter Estimation

The closed-form MLE estimates are:

$$\hat{\phi} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y^{(i)} = 1], \quad (1)$$

$$\hat{\boldsymbol{\mu}}_k = \frac{\sum_{i:y^{(i)}=k} \mathbf{x}^{(i)}}{\sum_{i:y^{(i)}=k} 1}, \quad k = 0, 1, \quad (2)$$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_{i=1}^N (\mathbf{x}^{(i)} - \boldsymbol{\mu}_{y^{(i)}})(\mathbf{x}^{(i)} - \boldsymbol{\mu}_{y^{(i)}})^\top. \quad (3)$$

3.3 Step 1 — Class Prior $\hat{\phi}$

Prior

There are $n_1 = 3$ admitted and $n_0 = 2$ rejected students out of $N = 5$:

$$\hat{\phi} = \frac{n_1}{N} = \frac{3}{5} = 0.6.$$

3.4 Step 2 — Class Means $\hat{\boldsymbol{\mu}}_0$ and $\hat{\boldsymbol{\mu}}_1$

Class Means

Class 1 (admitted: students 1, 2, 3):

$$\hat{\boldsymbol{\mu}}_1 = \frac{1}{3} \begin{pmatrix} 45 + 85 + 60 \\ 85 + 46 + 83 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 190 \\ 214 \end{pmatrix} = \begin{pmatrix} 190/3 \\ 214/3 \end{pmatrix} \approx \begin{pmatrix} 63.33 \\ 71.33 \end{pmatrix}.$$

Class 0 (rejected: students 4, 5):

$$\hat{\boldsymbol{\mu}}_0 = \frac{1}{2} \begin{pmatrix} 30 + 68 \\ 49 + 28 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 98 \\ 77 \end{pmatrix} = \begin{pmatrix} 49 \\ 38.5 \end{pmatrix}.$$

3.5 Step 3 — Pooled Covariance $\hat{\Sigma}$

Deviations from Class Means

Each example's deviation $\mathbf{x}^{(i)} - \boldsymbol{\mu}_{y^{(i)}}$:

i	$\mathbf{x}^{(i)}$	$y^{(i)}$	$\boldsymbol{\mu}_{y^{(i)}}$	$\mathbf{d}^{(i)} = \mathbf{x}^{(i)} - \boldsymbol{\mu}_{y^{(i)}}$
1	(45, 85)	1	(190/3, 214/3)	(-55/3, +41/3)
2	(85, 46)	1	(190/3, 214/3)	(+65/3, -76/3)
3	(60, 83)	1	(190/3, 214/3)	(-10/3, +35/3)
4	(30, 49)	0	(49, 38.5)	(-19, +10.5)
5	(68, 28)	0	(49, 38.5)	(+19, -10.5)

Outer Products

Compute $\mathbf{d}^{(i)}(\mathbf{d}^{(i)})^\top$ for each example:

Example $i = 1$: $\mathbf{d}^{(1)} = (-55/3, 41/3)^\top$

$$\begin{aligned}\mathbf{d}^{(1)}(\mathbf{d}^{(1)})^\top &= \frac{1}{9} \begin{pmatrix} 55^2 & -55 \cdot 41 \\ -55 \cdot 41 & 41^2 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 3025 & -2255 \\ -2255 & 1681 \end{pmatrix} \\ &\approx \begin{pmatrix} 336.11 & -250.56 \\ -250.56 & 186.78 \end{pmatrix}.\end{aligned}$$

Example $i = 2$: $\mathbf{d}^{(2)} = (65/3, -76/3)^\top$

$$\mathbf{d}^{(2)}(\mathbf{d}^{(2)})^\top = \frac{1}{9} \begin{pmatrix} 4225 & -4940 \\ -4940 & 5776 \end{pmatrix} \approx \begin{pmatrix} 469.44 & -548.89 \\ -548.89 & 641.78 \end{pmatrix}.$$

Example $i = 3$: $\mathbf{d}^{(3)} = (-10/3, 35/3)^\top$

$$\mathbf{d}^{(3)}(\mathbf{d}^{(3)})^\top = \frac{1}{9} \begin{pmatrix} 100 & -350 \\ -350 & 1225 \end{pmatrix} \approx \begin{pmatrix} 11.11 & -38.89 \\ -38.89 & 136.11 \end{pmatrix}.$$

Examples $i = 4, 5$: $\mathbf{d}^{(4)} = (-19, 10.5)^\top$ and $\mathbf{d}^{(5)} = (19, -10.5)^\top$ produce the same outer product:

$$\mathbf{d}^{(4)}(\mathbf{d}^{(4)})^\top = \mathbf{d}^{(5)}(\mathbf{d}^{(5)})^\top = \begin{pmatrix} 361 & -199.5 \\ -199.5 & 110.25 \end{pmatrix}.$$

Sum $S = \sum_{i=1}^N \mathbf{d}^{(i)}(\mathbf{d}^{(i)})^\top$:

$$\begin{aligned}S_{11} &= 336.11 + 469.44 + 11.11 + 361 + 361 = 1538.67, \\ S_{12} &= -250.56 - 548.89 - 38.89 - 199.5 - 199.5 = -1237.33, \\ S_{22} &= 186.78 + 641.78 + 136.11 + 110.25 + 110.25 = 1185.17.\end{aligned}$$

$$\hat{\Sigma} = \frac{S}{N} = \frac{1}{5} \begin{pmatrix} 1538.67 & -1237.33 \\ -1237.33 & 1185.17 \end{pmatrix} \approx \begin{pmatrix} 307.73 & -247.47 \\ -247.47 & 237.03 \end{pmatrix}.$$

The **negative** off-diagonal entry ($\hat{\Sigma}_{12} \approx -247.47$) reflects a negative correlation between exam scores within each class: a student who scores very high on one exam tends to score somewhat lower on the other, relative to their class mean.

3.6 Step 4 — Matrix Inverse $\hat{\Sigma}^{-1}$

For a 2×2 matrix $\Sigma = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, the inverse is $\Sigma^{-1} = \frac{1}{ad-b^2} \begin{pmatrix} d & -b \\ -b & a \end{pmatrix}$.

Inverting $\hat{\Sigma}$

With $a = 307.73$, $b = -247.47$, $d = 237.03$:

$$\det(\hat{\Sigma}) = ad - b^2 = 307.73 \times 237.03 - (-247.47)^2 = 72943.1 - 61241.5 = 11701.6.$$

$$\hat{\Sigma}^{-1} = \frac{1}{11701.6} \begin{pmatrix} 237.03 & 247.47 \\ 247.47 & 307.73 \end{pmatrix} \approx \begin{pmatrix} 0.02025 & 0.02115 \\ 0.02115 & 0.02629 \end{pmatrix}.$$

3.7 Step 5 — GDA Decision Boundary Parameters

Setting $P(y = 1 | \mathbf{x}) = P(y = 0 | \mathbf{x})$ and simplifying (the shared Σ causes the quadratic $\mathbf{x}^\top \hat{\Sigma}^{-1} \mathbf{x}$ terms to cancel), the decision boundary is the hyperplane $\boldsymbol{\theta}^\top \mathbf{x} + \theta_0 = 0$ where:

$$\boldsymbol{\theta} = \hat{\Sigma}^{-1}(\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0),$$

$$\theta_0 = -\frac{1}{2} \hat{\boldsymbol{\mu}}_1^\top \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}_1 + \frac{1}{2} \hat{\boldsymbol{\mu}}_0^\top \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}_0 + \log \frac{\hat{\phi}}{1 - \hat{\phi}}.$$

Computing $\boldsymbol{\theta}$

$$\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0 = \begin{pmatrix} 190/3 - 49 \\ 214/3 - 38.5 \end{pmatrix} = \begin{pmatrix} 43/3 \\ 197/6 \end{pmatrix} \approx \begin{pmatrix} 14.33 \\ 32.83 \end{pmatrix}.$$

$$\boldsymbol{\theta} = \hat{\Sigma}^{-1} \begin{pmatrix} 14.33 \\ 32.83 \end{pmatrix} = \begin{pmatrix} 0.02025 \times 14.33 + 0.02115 \times 32.83 \\ 0.02115 \times 14.33 + 0.02629 \times 32.83 \end{pmatrix} = \begin{pmatrix} 0.9846 \\ 1.1664 \end{pmatrix}.$$

Computing θ_0

$$\hat{\boldsymbol{\mu}}_1^\top \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}_1 = (63.33, 71.33) \hat{\Sigma}^{-1} (63.33, 71.33)^\top \approx 468.70,$$

$$\hat{\boldsymbol{\mu}}_0^\top \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}_0 = (49, 38.5) \hat{\Sigma}^{-1} (49, 38.5)^\top \approx 109.85,$$

$$\log \frac{\hat{\phi}}{1 - \hat{\phi}} = \log \frac{0.6}{0.4} = \log \frac{3}{2} \approx 0.4055.$$

$$\theta_0 = -\frac{1}{2}(468.70) + \frac{1}{2}(109.85) + 0.4055 = -234.35 + 54.93 + 0.41 \approx -118.95.$$

$$\boldsymbol{\theta}^\top \mathbf{x} + \theta_0 = 0 \iff 0.9846 x_1 + 1.1664 x_2 - 118.95 = 0.$$

This is a **linear** boundary — a direct consequence of the shared covariance $\hat{\Sigma}$.

3.8 Step 6 — Prediction for $\mathbf{x}^* = (45, 85)^\top$

Score (log-posterior ratio):

$$\boldsymbol{\theta}^\top \mathbf{x}^* + \theta_0 = 0.9846 \times 45 + 1.1664 \times 85 - 118.95 = 44.31 + 99.14 - 118.95 = 24.50.$$

Since the score $\gg 0$: $\hat{y} = 1$ (**Admitted**).

The GDA model is *highly* confident — the test point $(45, 85)$ lies very close to the class-1 cluster ($\hat{\boldsymbol{\mu}}_1 \approx (63.33, 71.33)$) and far from the class-0 cluster ($\hat{\boldsymbol{\mu}}_0 = (49, 38.5)$).

3.9 GDA Posteriors on the Training Set

Applying the GDA classifier to all training points:

i	x_1	x_2	y	Score	$P(y = 1 \mid \mathbf{x})$	$\hat{y}$
1	45	85	1	+24.50	≈ 1.000	1 ✓
2	85	46	1	+18.39	≈ 1.000	1 ✓
3	60	83	1	+36.94	≈ 1.000	1 ✓
4	30	49	0	-32.26	≈ 0.000	0 ✓
5	68	28	0	-19.34	≈ 0.000	0 ✓

All 5 training examples are classified correctly (training accuracy = 100%).

4 Connection Between GDA and Logistic Regression

Key Theorem

If the GDA assumptions hold (Gaussian class-conditionals with shared covariance), then the posterior

$$P(y = 1 \mid \mathbf{x}) = \frac{P(\mathbf{x} \mid y = 1) P(y = 1)}{P(\mathbf{x} \mid y = 1) P(y = 1) + P(\mathbf{x} \mid y = 0) P(y = 0)}$$

simplifies exactly to the logistic sigmoid form:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\boldsymbol{\theta}^\top \mathbf{x} + \theta_0) = \frac{1}{1 + e^{-(\boldsymbol{\theta}^\top \mathbf{x} + \theta_0)}}.$$

Why? After substituting the Gaussian densities into the ratio, the quadratic $\mathbf{x}^\top \hat{\Sigma}^{-1} \mathbf{x}$ terms appear identically in both numerator and denominator (because Σ is shared) and *cancel exactly*, leaving only a linear function of $\mathbf{x}$ in the exponent.

This means **GDA with shared covariance $\Rightarrow$ logistic regression model**. The converse does not hold: logistic regression does not require the class-conditionals to be Gaussian. GDA makes a strictly stronger assumption, but in return obtains a closed-form solution in one pass over the data.

5 Comparison: GDA vs. Logistic Regression on This Problem

Both classifiers agree on the test point, predicting **Admitted**. Their decision boundaries differ because:

- Logistic regression finds the boundary that best separates the classes *without any distributional assumption*, fitting directly to the labels.
- GDA places the boundary according to its model of the two Gaussian clouds. Its stronger assumption about the data-generating process yields a different (and on this tiny dataset, more extreme) posterior.

6 Summary of All Computed Parameters

Logistic Regression (MLE at convergence):

$$\hat{\boldsymbol{\theta}} = \begin{pmatrix} -25.16 \\ 0.206 \\ 0.201 \end{pmatrix}, \quad \text{decision boundary: } -25.16 + 0.206 x_1 + 0.201 x_2 = 0.$$

GDA (closed-form):

$$\hat{\phi} = 0.6, \quad \hat{\boldsymbol{\mu}}_0 = \begin{pmatrix} 49.00 \\ 38.50 \end{pmatrix}, \quad \hat{\boldsymbol{\mu}}_1 = \begin{pmatrix} 63.33 \\ 71.33 \end{pmatrix},$$

$$\hat{\boldsymbol{\Sigma}} = \begin{pmatrix} 307.73 & -247.47 \\ -247.47 & 237.03 \end{pmatrix}, \quad \hat{\boldsymbol{\Sigma}}^{-1} \approx \begin{pmatrix} 0.02025 & 0.02115 \\ 0.02115 & 0.02629 \end{pmatrix}.$$

$$\boldsymbol{\theta}_{\text{GDA}} = \begin{pmatrix} 0.9846 \\ 1.1664 \end{pmatrix}, \quad \theta_{0,\text{GDA}} = -118.95.$$

Decision boundary: $0.985 x_1 + 1.166 x_2 = 118.95$.

Test point $\mathbf{x}^* = (45, 85)^\top$:

$$\hat{y}_{\text{LR}} = 1 \ (P \approx 0.768), \quad \hat{y}_{\text{GDA}} = 1 \ (P \approx 1.000).$$

Both models predict: **Admitted.**

7 GDA vs. Logistic Regression: Key Differences

7.1 Side-by-Side Comparison

GDA vs. Logistic Regression

Aspect	GDA	Logistic Regression
Type	Generative	Discriminative
What is modelled?	$P(\mathbf{x} y)$ and $P(y)$	$P(y \mathbf{x})$ directly
Assumptions	Gaussian class-conditionals, shared Σ	No distributional assumption on $\mathbf{x}$
Parameters	$\phi, \mu_0, \mu_1, \Sigma$	$\theta \in \mathbb{R}^{d+1}$
Estimation	Closed-form MLE (one pass)	Gradient descent (iterative)
Decision boundary	Linear (when Σ shared)	Linear
Data efficiency	Better when assumptions hold	Better when assumptions violated
Works with non-Gaussian?	Poorly	Yes

7.2 When GDA Wins

Use GDA when...

- Your features are **continuous and approximately Gaussian** (or can be transformed to be so). This is common in many scientific domains.
- You have **limited training data**. GDA's strong assumptions act as a form of regularisation — you need fewer examples to get good estimates because the model “knows” the data is Gaussian.
- Training speed matters: GDA requires **only one pass** over the data with no iterative optimisation.
- You need to **generate** new examples or detect out-of-distribution inputs (since you have an explicit model of $P(\mathbf{x})$).

7.3 When Logistic Regression Wins

Use Logistic Regression when...

- Your features are **not Gaussian** — binary features, counts, skewed distributions.
- You have **lots of data** and want the most accurate model without assumptions.
- Your features include **discrete or categorical variables** (e.g., spam detection: word presence/absence).